Kishan Prajapati

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Education

Northeastern University · M.S. in Computer Science · GPA 3.78/4.00 Jan 2026 – May 2028

P P Savani University · B.Tech. in Information Technology and Engineering Oct 2021 – May 2025

Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash
ML and LLMs: PyTorch, Hugging Face, PEFT (QLoRA, LoRA), scikit-learn, embeddings, RAG, LLM evaluation
Backend and Frontend: FastAPI, Node.js, Express, REST APIs, WebSockets, React, Vite, Tailwind CSS
Data and Infrastructure: PostgreSQL (pgvector, HNSW), MongoDB, Docker, GitHub Actions (CI/CD), Linux, AWS
Testing and Design: pytest, JUnit, PIT mutation testing, TDD, object-oriented design (MVC, SOLID)

Experience

Student Researcher (First Author), Northeastern University, Prof. Malihe Alikhani Jan 2026 – Present
LLMs for clinical text · PyTorch, Hugging Face, QLoRA, A100 GPUs · manuscript in preparation · case study · code

- Fine-tuned Llama 3.1 8B with QLoRA to simplify clinical text; MedLane SARI rose from **16.6 to 55.5**.
- Changing only the prompt, with model weights fixed, raised retention of key medical terms from 0.15 to 0.63 F1.
- Built a reproducible train/eval harness for Llama, BioBART, and FLAN-T5 across 28 model-dataset conditions.
- Reported that fine-tuning gains did not transfer to an unseen dataset (disjoint 95% confidence intervals).

Software Developer Intern, Technonite (software services firm) Dec 2024 – Mar 2025
Node.js, Express, React, MongoDB, WebSockets, Google Maps API *Surat, India*

- Designed REST APIs and MongoDB schemas for 3 live client products, shipped through each client's release process.
- Developed a real-time dashboard streaming live vehicle positions to a logistics client's dispatchers over WebSockets.
- Created a reusable React component library adopted across all 3 client apps, replacing duplicated interface code.

Graduate Teaching Assistant, Northeastern University, Khoury College May 2026 – Present
CS 3000 Algorithms and Data, DS 3000 Foundations of Data Science, CS 1800 Discrete Structures *Boston, MA*

- Teach algorithms, data structures, and complexity analysis to 500+ students in recitations and office hours.
- Review student designs for correctness and time complexity; flagged an NP-hard formulation before a team built it.
- Wrote least-squares and eigenvalue study guides for DS 3000, checking every worked example against NumPy.

Undergraduate Researcher, P P Savani University, Dr. Jayashri Patil Nov 2023 – Mar 2025
Depression detection from social media text · Python, scikit-learn, TF-IDF, LIWC *Surat, India*

- Compared Naive Bayes, logistic regression, and SVM on 20,000 labeled tweets; best accuracy 88.1% (ICICC 2024).
- Second author of a 79-reference review of AI for depression detection (CML 2025).
- Cleaned and hand-labeled a 49,000-row text dataset spanning 490+ life-event categories and 17 emotion labels.

Projects

Sieve: Hybrid Search Engine over 182,853 Research Papers · Solo · getsieve.vercel.app Jul 2026 – Present
Python, FastAPI, PostgreSQL/pgvector, ONNX Runtime, React/TypeScript, Docker · 254 tests in CI · case study

- Shipped the full stack solo, combining keyword and vector search with rank fusion, plus accounts and team screening.
- Embedded every paper with a 33M-parameter model on an 8 GB laptop with no GPU, peaking at 2.05 GB of memory.
- Paired timing put approximate (HNSW) search at **7.7x** the speed of an exact scan (95% CI 7.5–8.0), 0.978 recall@20.
- Used EXPLAIN ANALYZE to find a missing index, cutting a hot-path query from 1,140 ms to under 1 ms.
- Made ingestion crash-safe on an idempotent job queue; a worker killed with SIGKILL resumed with zero gaps.

Calendar: Java Scheduling App with CLI and GUI · Team of 2 2026

- Architected one MVC model, following SOLID principles, behind a CLI, a headless scripting mode, and a Swing GUI.
- Extended an unfamiliar codebase with recurring events and time zones without rewriting its core design.
- Backed every change with JUnit tests, JaCoCo coverage, and PIT mutation testing to confirm tests catch real faults.